Correlation in R

PS50008A Week 15 | Lab Walkthrough

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Welcome to Lab

Correlation in R

Today's Focus

Analysing relationships between variables. Today you will:

- Create scatterplots to visualise relationships
- Calculate Pearson's r and Spearman's rho
- Test correlations for significance
- Report correlations in APA format

Lab Structure (~90 min)

Time	Activity
10:00-10:10	These slides + setup
10:10-10:30	Part 1: Scatterplots
10:30-10:50	Part 2: Interpreting correlations
10:50-11:10	Part 3: Calculating r in R
11:10-11:30	**Research Plan work time**
11:30-12:00	Research Plan Q&A + wrap-up

⚠ **Warning**

Important: Reserve at least 30 min for Research Plan work!

Open the Activity

Navigate to: **Week 15 > Lab > Activity**

Make sure WebR loads before proceeding.

NEW: Correlation Playground

> **Tip**
>
> **Interactive Shiny App** for exploring correlation
>
> littlemonkeylab.github.io/GS_shinyapps/apps/correlation-playground/

Students can:

- Adjust correlation strength with sliders
- See how r values translate to scatterplot patterns
- Build intuition before calculating

Recommend using this during Part 1 & 2!

Last Week Before Reading Week

Note

No labs next week (Reading Week: 16-20 February). Use the time to consolidate correlation and prepare for Week 16.

Part 1: Scatterplots

Visualising Relationships (~25 min)

Why Start with Scatterplots?

Always visualise before calculating!

Scatterplots reveal:

- Direction (positive or negative)
- Strength (tight cluster or spread out)
- Linearity (straight line or curved)
- Outliers (unusual points)

Creating Scatterplots in R

```r
ggplot(data, aes(x = var1, y = var2)) +
  geom_point() +
  geom_smooth(method = "lm")
```

The `geom_smooth(method = "lm")` adds the line of best fit.

What to Look For

Ask students to describe:

1. **Direction**: Does it go up or down?
2. **Strength**: How tightly clustered?
3. **Outliers**: Any unusual points?
4. **Linearity**: Is a straight line appropriate?

Part 2: Calculating Correlations

Pearson’s r and Spearman’s rho (~30 min)

Pearson's r

Use Pearson's r when:

- Both variables are continuous
- Relationship is approximately linear
- No extreme outliers

```r
cor.test(x, y, method = "pearson")
```

Spearman’s rho

Use Spearman’s rho when:

- Data are ordinal (ranks)
- Relationship is monotonic but not linear
- Outliers are a concern

```r
cor.test(x, y, method = "spearman")
```

Reading the Output

Output	Meaning
cor	The correlation coefficient
t	Test statistic
df	Degrees of freedom (n - 2)
p-value	Is it significant?

Interpreting Strength

r value	Interpretation
.10 - .29	Weak
.30 - .49	Moderate
.50+	Strong

Remind students: these are guidelines, not strict rules.

Part 3: Correlation Matrices

Multiple Variables (~25 min)

When You Have Many Variables

> **Correlation matrix** shows correlations between ALL pairs of variables at once.
>
> Useful for exploring DataLab data with multiple measures.

The R Code

```r
# Select numeric variables
cor_data <- select(data, var1, var2, var3)

# Calculate correlation matrix
cor(cor_data, use = "complete.obs")
```

Reading a Correlation Matrix

Key points:

- Diagonal is always `1.00` (variable with itself)
- Matrix is symmetric (top-right = bottom-left)
- Look for strong correlations (> `.50` or < `-.50`)

Part 4: APA Reporting

Writing Up Correlations (~20 min)

APA Format

> **Template:** r(df) = .XX, p = .XXX
>
> **Example:** r(48) = .42, p = .003

Note: df = n - 2 for correlations.

Example Write-Up

> "There was a significant positive correlation between confidence and accuracy, $r(48) = .42$, $p = .003$. Participants who reported higher confidence tended to perform better on the memory task."

Correlation vs Causation

Key Point

Correlation does NOT imply causation!

Students must avoid causal language:

- NO: “Confidence causes better performance”
- YES: “Confidence was associated with better performance”

Research Plan Work Time

Use Remaining Lab Time Productively

Research Plan Reminder

Due Friday 27 February, 4pm via Learn

This is the **last lab before Reading Week** - encourage students to use this time wisely!

What Students Should Work On

If they haven't started:

- Choose research question from approved topics
- Identify IV and DV clearly
- Decide between/within-subjects design

If already started:

- Refine hypotheses (H1 and H0)
- Check operationalisation is measurable
- Review marking criteria

Circulate and Offer Help

Good questions to ask students:

- “What’s your IV? How will you manipulate/measure it?”
- “Is your hypothesis specific and testable?”
- “What design are you using? Why?”
- “Have you thought about confounding variables?”

Common Issues to Watch For

Problem	Guidance
Vague hypothesis	Must be specific and directional
Unmeasurable DV	How exactly will you measure it?
Wrong design	Check if same or different participants
Causal language	Remind: only experiments show causation

Wrap-Up

Key Takeaways

Today We Learned

1. **Always visualise first** with scatterplots
2. **Pearson's r** for linear relationships between continuous variables
3. **Spearman's rho** for ordinal data or non-linear relationships
4. **Correlation does not equal causation**

Next Steps

Direct students to:

1. Complete the **Week 15 Self-Assessment Quiz**
2. **Reading Week**: Review correlation, prepare for regression
3. **Week 16**: Project Groups - Literature Review session

Assessment Reminder

⚠ Research Plan Due Week 16

Due **Friday 27 February** at 4pm via Learn. Use Reading Week to finalise your submission.

Questions?

Then let’s begin the activity!